PSO for Optimizing Topology Design of Neural Networks

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Abstract

Designing effective neural network architectures typically requires extensive manual tuning of hyperparameters and structural choices. This project explores the use of Particle Swarm Optimization (PSO) to automatically search for high-performing neural network architectures. The objective is to determine whether PSO can efficiently discover architectures that achieve strong predictive performance while maintaining reasonable model complexity and training cost.

Each particle in the swarm represents a candidate architecture encoded as a vector of design parameters such as number of hidden layers and learning rate. Candidate networks are evaluated on datasets from MedMNIST [4][5], using a multi-objective fitness function that considers validation accuracy, model size, and training time. To address the computational cost of training many candidate networks, strategies for reducing evaluation overhead are explored.

The project analyzes how neural network architectures evolve during optimization, and examines swarm diversity and convergence behavior. Expected outcomes include identifying architecture patterns associated with strong performance and demonstrating whether PSO provides a more efficient search strategy than baseline methods.

1 Project Description

1.1 Motivation

The manual tuning of neural network hyperparameters and topology design is a computationally expensive and laborious task. The limitations of hand-tuning neural networks has inspired automated architecture engineering processes known as Neural Architecture Search (NAS) [1][2][3]. NAS can be employed using various machine learning paradigms to automatically optimize hyperparameters. NAS methods are already outperforming manually designed architectures in several machine learning and deep learning domains.

1.2 Proposed Method

This project proposes to employ particle swarm optimization (PSO) as a method of NAS on the MedMNIST dataset [6][7]. Each particle will encode a possible configuration of the neural network's architecture, in the form of a vector. Particle fitness is determined by validation accuracy, model size and training time of the network using the particle's encoded architecture. The PSO algorithm will manipulate the particles to uncover the vector configuration that best optimizes the objectives.

1.3 Experimental Setup

The hyperparameters and topology choices being encoded are hidden layer size, number of hidden layers, learning rate, momentum, number of epochs, batch size and dropout rate. Every particle will thus be a 7-dimensional vector. An lbest implementation of PSO with dynamic inertia weight will be used, along with optimal values for the cognitive and social components to ensure convergence.

1.4 Planned Analysis

The best hyperparameters found by PSO will be compared to randomly and manually tuned hyperparameters to determine if automatic hyperparameter tuning results in a significant improvement of the objectives. The results from PSO will be analyzed further to understand the diversity and convergence characteristics of the particles.

Analyses will also be performed to determine which architecture choices resulted in the biggest improvement of the objective functions.

1.4.1 Visualizations

Some visualizations we plan to create and analyze include:

- Line plots displaying best fitness over iterations, indicating how the best architecture improves during optimization.
- Line plots showing average swarm fitness over iterations, showing whether the entire swarm improves or if only a few particles do.
- Line plots for swarm diversity over time, showing exploration versus exploitation behavior of the swarm during optimization.
- Architecture parameter distributions, either histograms or boxplots, for parameters such as learning rate, batch size, and number of layers.
- A Pareto front diagram, visualizing the trade-off in our multi-objective optimization between accuracy and computational cost.
- A bar chart comparing PSO versus other search methods (random, manual tuning).

1.5 Expected Outcomes

The primary expected outcome of this project is the development of a neural architecture search method based on PSO that can effectively and efficiently identify optimal neural network architectures. The project is expected to provide insights into which architectural and training parameter configurations consistently lead to strong performance across datasets from MedMNIST.

References

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